Ancient methods
Egyptians had developed a glass-casting method similar to Vitraglyphic thousands of years ago in which finely crushed glass was mixed with a binding material such as gum arabic and water, deposited on a negative mould to form a coating

Movable type
Bing Shen(China) is often credited for having invented the movable type printing technology much before Gutenberg

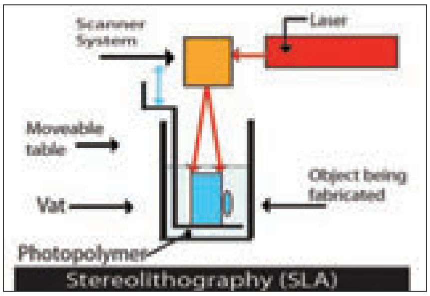
acrylics are more commonly used.

5. Clean up the final product and finish. This may involve removal of support structures, curing, sanding, blowing or breaking away excess material or painting the finished product.

The process is quicker, cheaper and more precise than the earlier developmental process. The similarity in process, however, ends here. There are several major variations that are in vogue (more than six are easily available today), so let's take a good look at some of them and how they work.

Charles Hull was the first to shake the world awake to the idea of 3D printing with his patented process of 'Stereolithography' (Greek for 'solid-stone-writing') in 1986. The machine primarily consists of an ultraviolet laser, a set of mirrors (X/Y) and a platform immersed in a vat full of liquid photopolymer – a fluid plastic. Once the machine gets going, the ultraviolet laser (bouncing off the mirrors) 'draws' on the polymer that is on the platform. The material, being sensitive to light hardens into a flat rigid layer when exposed to the laser. This becomes the bottom of the prototype. The thickness of the layer is usually about 0.004 to 0.006 of an inch but you could manipulate this as per your requirements. The platform then descends into the tank and the polymer that spreads over it is then given the same treatment by the laser, hardening the next cross section of the model. While there are plenty of resins that could be used, they're usually the ABS (acrylonitrile-butadiene-

Charles Hull's breakthrough invention



What lies beneath

styrene) kind or moulded polypropylene – clear polymers that can withstand very high temperatures.

Finally, the solid object is taken out of the tank, the extra polymer washed off and the support portions broken off and put into an ultraviolet oven to be completely cured. However, the size of the model you make must be restricted to the size of the tank and the build platform. To make very large objects, typically, the object must be divided into parts which must be put together later.

Selective Laser Sintering (SLS)

Sintering is, basically, the process of heating a powdered material in a sintering furnace to a little below its melting point so that the constituents stick to one another and an object forms. Carl Deckard patented the Selective Laser Sintering technology in 1989 based on his master's thesis at the University of Texas. Like SLA, SLS uses a laser beam – but instead of working to cure a liquid polymer, it uses materials like nylon, metal (alloys), elastomer, thermoplastics (filled with either glass, aluminium or polyamide) etc., in a powdered form.

The platform rests on a bin of the powder. An extremely hot laser draws the two-dimensional bitmap on the first layer of the powder that's on the build platform. As the laser moves over the powder, the particles fuse to form a solid layer. A roller moves over it to level it flat. The platform is then lowered, the powder deposited on the first layer and the process repeated to make a new layer on top of the first. The technique has several advantages when it comes to building smaller models and companies like EOS have exploited these to make dental implants like crowns and bridges.

Sticky sinters

The powder that is in excess - or has failed to sinter - functions as a support material during the process and can be brushed or blown off later.

Fused Deposition Modelling

FDM was developed by S. Scott Crump, this technology came into the limelight in the 1990s. The unique thing about this technique is that the building material is actually a thin solid filament to begin with. The solid material (usually ABS, polycarbonates, polyphenylsufones, or waxes) is extruded from a spool to a movable head that can be manipulated by servomotors (and this is probably the only thing about 3D printing that can justify

Today's FDS printers are aesthetically designed

the use of the word 'print' when we aren't really printing anything at all). Another secondary nozzle squeezes out a strand of material which could serve as a support structure for the model. The temperature of the platform is much lower than the material's melting point so it hardens on